

4 Tuberculosis (TB) is an infectious disease that is caused by a bacterial pathogen.

The pathogen has mechanisms to avoid digestion by phagocytes. Macrophages may engulf the bacteria, but in some cases the bacteria remain alive within the cells instead of being killed.

- (a) Although there are two main species of bacterium causing TB, it is rare for the species that causes bovine TB in cattle to infect humans.

Name the species of bacterium that is the **main** cause of TB in humans.

..... [1]

- (b) Explain how the pathogen named in (a) is transmitted from a person with the disease to a person who is uninfected.

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.....
.....
.....
..... [2]

- (c) State the type of therapeutic drug used to treat TB.

..... [1]

- (d) Fig. 4.1 is a photomicrograph of a section of lung tissue taken from a person who has **not** been infected with the bacterial pathogen and who does **not** have TB.

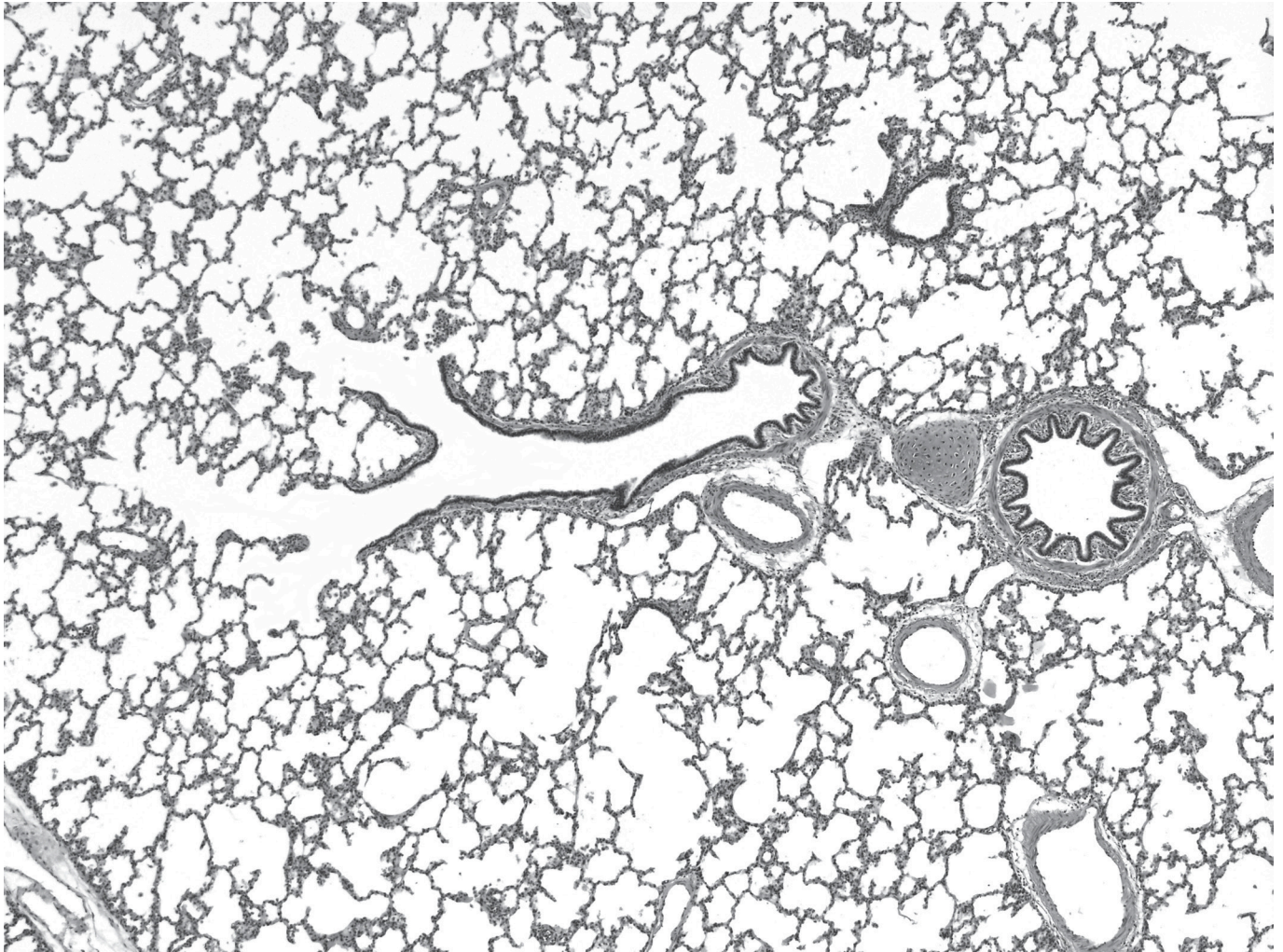


Fig. 4.1

Blood vessels and some structures of the gas exchange system are visible in Fig. 4.1.

On Fig. 4.1, use a label line and label:

- a bronchus
- a bronchiole
- a blood vessel.

[3]

[illegible]

Fig. 4.2 is a photomicrograph of a granuloma in lung tissue.

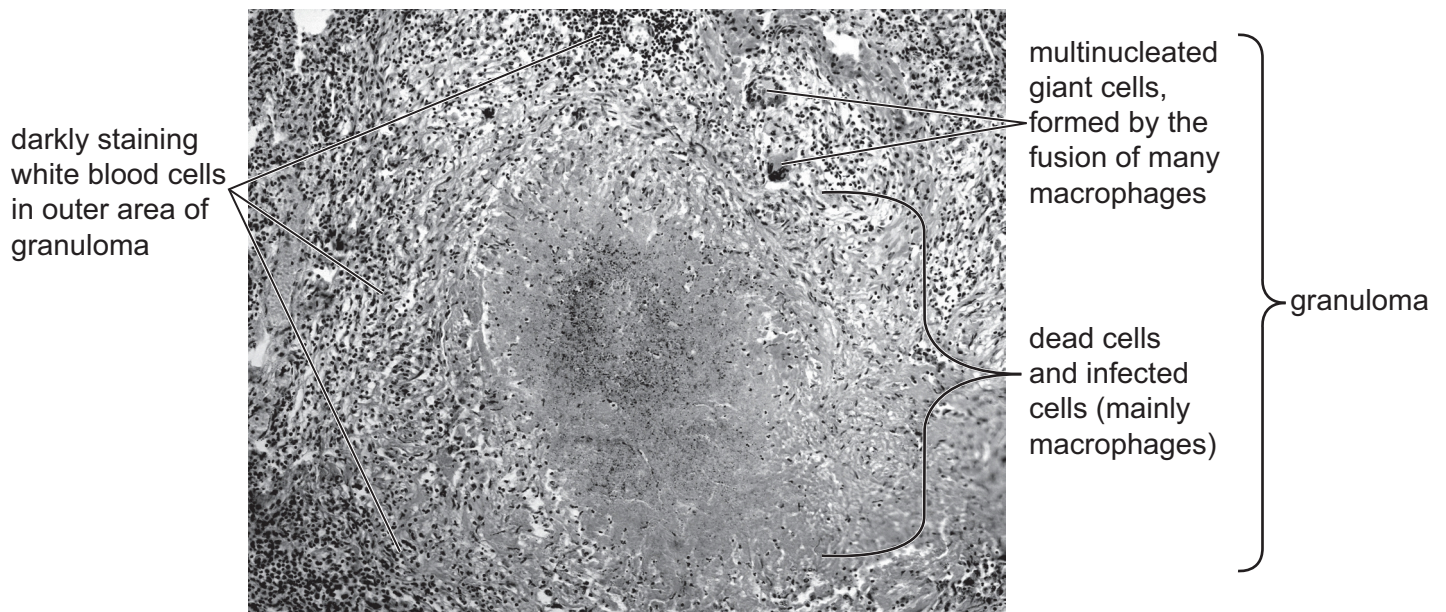


Fig. 4.2

With reference to Fig. 4.1 and Fig. 4.2, describe **and** explain how the changes that occur as a result of granuloma formation:

- can affect gas exchange and harm the health of an infected person
- may help to prevent TB developing in other parts of the body.

..... [5]

[Total: 12]